

# COSIMA Cookbook: a computational learning module for analysing ocean and sea-ice model output

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DOI: [10.xxxxxx/draft](https://doi.org/10.xxxxxx/draft)

## Software

- [Review](#)
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Submitted: 01 January 1970

Published: unpublished

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## Summary

The COSIMA Cookbook is an open computational learning module for analysing ocean and sea-ice model output in Jupyter notebooks. It has been developed by the Consortium for Ocean-Sea Ice Modelling in Australia (COSIMA; <https://cosima.org.au>) as a community resource for researchers, students, and practitioners working with large gridded datasets, especially output from the ACCESS ocean-sea ice model configurations (e.g. Kiss et al., 2020). The repository combines introductory tutorials with worked analysis examples, all exposed through a browsable Sphinx documentation site (Figure 1) and backed by a citable archived release (Constantinou et al., 2026).

## COSIMA Cookbook

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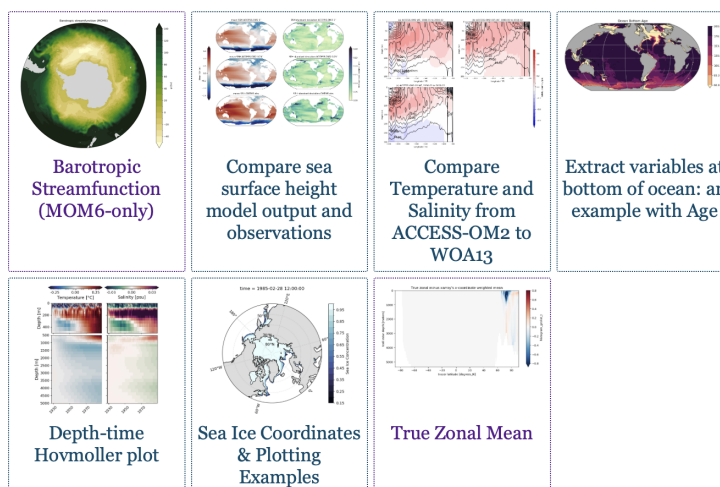
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## Appetisers (easy)



latest

**Figure 1:** The COSIMA Cookbook documentation website, showing the browsable Sphinx-based landing page used to navigate tutorials and recipes. The live site is available at <https://cosima-recipes.readthedocs.io>.

Following the “Cookbook” concept, the website sections are deliberately named using a gastronomy theme. “Cooking Tutorials” refers to tutorials that teach generic, transferable techniques for handling ocean–sea ice model output. Then comes the main part of any cookbook, its recipes. Here, by “recipes” we mean self-contained notebooks showing how to perform concrete diagnostics and analyses on ocean and sea-ice datasets. First come the “Easy Recipes” that provide a good entry point after the tutorials; then the “Advanced Recipes” that are more elaborate and advanced analysis examples, and last the “Regional Specialties” contains recipes for regional model configurations (Barnes et al., 2024). Together these sections present the documentation as a browsable collection of lessons and recipes gathered into a single cookbook.

Pedagogy and instructional structure are central to the project. The idea is that these recipes provide the starting point for more elaborate analyses. “Cooking Tutorials” introduces generic skills such as loading model output, working with labelled arrays, plotting, and interacting with shared data catalogues. These tutorials lead into domain-focused recipes giving learners an incremental path from first contact with the data ecosystem to adaptation of full workflows for their own science questions.

Within this structure, the tutorials include examples of loading model output through Intake-based catalogues (Intake developers, 2026) and incorporating Cartopy (Elson et al., 2024) for map-based visualisation. The recipe collection also includes a Lagrangian particle-tracking workflow built with Parcels (Delandmeter & Seville, 2019), as well as analyses that use xarray (Hoyer & Hamman, 2017) and Dask (Dask Development Team, 2016; Rocklin, 2015) to process very large multi-dimensional datasets in parallel, xgcm

(Abernathy et al., 2022) to handle variables defined on staggered finite-volume grids and xESMF (Zhuang et al., 2025) for regridding model output.

The Cookbook grew from a practical need inside the COSIMA community: the tools used to analyse modern ocean model output are powerful, but the gap between package-level documentation and a reproducible end-to-end workflow remains large. Users need to understand not only Python and Jupyter (Kluyver et al., 2016), but also how to navigate high-dimensional model output, shared high-performance computing environments, and domain-specific analysis conventions. The Cookbook addresses that gap by packaging reusable workflows in the same medium in which users actually work.

## Statement of Need

Ocean and climate model analysis has a steep entry cost. New users must learn how to find datasets, load them efficiently, interpret metadata, operate on multi-dimensional arrays, and produce scientifically meaningful diagnostics. General-purpose libraries such as xarray (Hoyer & Hamman, 2017) provide the foundations for these tasks, but they do not by themselves show a learner how to translate a research question into a robust analysis workflow for a specific model and computing environment.

That gap is not only pedagogical; it directly affects how quickly model output can be turned into science. For a beginner, becoming productive with a model often requires substantial time just to understand the structure of the output, the relevant conventions, and the computational environment in which analysis is expected to run. There is also substantial scientific labour embedded in many common diagnostics. Some calculations, such as cross-contour transport, require non-trivial development and validation before they can be used confidently. If every new user or project had to reconstruct those workflows independently, a large amount of effort would be repeatedly spent on rebuilding analysis code rather than doing science with the model.

The COSIMA Cookbook fills that need with a domain-specific, openly maintained collection of computational lessons and examples. Its contribution is not a new analysis library; rather, it is a reusable scientific and educational resource that exposes complete, well-documented workflows in the same notebook format in which many researchers actually explore data and communicate methods. By choosing notebooks instead of packaging these examples only as a software library, the project makes intermediate reasoning, methodological choices, and practical execution details visible to learners while still giving more experienced users analysis patterns they can adapt directly for research. This aligns well with JOSE's emphasis on open educational resources that enable computational learning through authentic practice.

Several design decisions make the Cookbook particularly useful for adoption by other groups. First, each notebook is intended to be self-contained and well-documented, so learners can read, run, and modify a complete workflow without reconstructing missing context from scattered notes. Second, the repository distinguishes tutorials from recipes. Tutorials teach transferable skills, while recipes demonstrate how those skills combine in realistic analysis tasks. Third, the project includes contributor guidance and notebook review conventions that encourage new analyses to be written in a reusable and pedagogically clear style. A caveat is that some recipes use high-resolution model output with datasets of order terabytes, so the underlying data are not always practical to distribute or mirror in full. However, beyond the initial data-loading step, which usually occurs early in each recipe, the analysis workflows are generally applicable to most operational ocean-sea ice model outputs that can be loaded with xarray (Hoyer & Hamman, 2017).

Thus, the structure is valuable beyond the immediate COSIMA context. Many research communities maintain model output on shared infrastructure and face the same challenge: turning expert tacit knowledge into examples that newcomers can adapt. The Cookbook

106 offers a model for how a scientific collaboration can capture that knowledge in version-  
107 controlled notebooks, publish it as a living resource, and continuously improve it through  
108 community contribution.

## 109 Educational Design and Experience of Use

110 The learning experience is organised as a progression. The documentation points new users  
111 first to introductory lessons, including material on loading, slicing, and visualising model  
112 output. Learners then move to more advanced tutorials and finally to recipe notebooks  
113 that address concrete scientific questions. The categories of “easy” and “advanced” recipes  
114 provide a lightweight pedagogical cue about expected complexity and scope, with “regional  
115 specialties” specifically covering recipes for regional model configurations (Barnes et al.,  
116 2024).

117 The intended mode of use is also explicit. The repository is designed around Jupyter-  
118 based analysis on the Australian National Computational Infrastructure, with guidance  
119 on running notebooks through ARE/JupyterLab sessions and on accessing shared data  
120 holdings. This makes the Cookbook more than a static collection of examples: it is  
121 operational documentation for a real analysis environment. At the same time, the  
122 notebooks demonstrate broadly transferable patterns for working with labelled geophysical  
123 data, so individual lessons can be reused or adapted outside that environment.

124 The project also supports community authorship. Contributors are encouraged to submit  
125 new notebooks through pull requests, document their methods clearly, and generalise  
126 workflows so that they remain useful to others. In that sense, the Cookbook functions  
127 both as a learning module for end users and as a framework for teaching reproducible  
128 scientific communication through notebook design and peer review. The easy access to  
129 data and analysis provided by the COSIMA Cookbook showcases best practices to the  
130 broader oceanographic community and enabled rapid community adoption of the ACCESS  
131 ocean and sea-ice model configurations internationally, facilitating to date well over 100  
132 peer-reviewed papers and scores of PhD projects.

## 133 Author order

134 The first two authors contributed equally; authors from the third position onward are  
135 listed alphabetically by surname.

## 136 Acknowledgements

137 We are grateful to the COSIMA community for developing and maintaining the model  
138 output, workflows, and pedagogical practices and also for the great community spirit  
139 that defines COSIMA. We acknowledge the National Computational Infrastructure (NCI),  
140 which is supported by the Commonwealth government of Australia for the computational  
141 resources provided under the National Computational Merit Allocation Scheme and the  
142 Australian National University Merit Allocation Scheme. We further acknowledge the  
143 ACCESS National Research Infrastructure (ACCESS-NRI) ecosystem that make the  
144 operational use of these notebooks possible. ACCESS-NRI’s ACCESS-OM2 and ACCESS-  
145 OM3 models are enabled by the Australian Government’s National Collaborative Research  
146 Infrastructure Strategy (a.k.a. NCRIS). We acknowledge funding from the Australian  
147 Research Council under the Centre of Excellence for the Weather of the 21st Century  
148 CE230100012, the Linkage Infrastructure, Equipment and Facilities LP160100073 and  
149 LP200100406, and the Discovery Project DP240101274.

## References

- Abernathy, R. P., Busecke, J. J. M., Smith, T. A., Deauna, J. D., Banihirwe, A., Nicholas, T., Fernandes, F., James, B., Dussin, R., Cherian, D. A., Caneill, R., Sinha, A., Uieda, L., Rath, W., Balwada, D., Constantinou, N. C., Ponte, A., Zhou, Y., Uchida, T., & Thielen, J. (2022). *xgcm* (Version v0.8.1). Zenodo. <https://doi.org/10.5281/zenodo.7348619>
- Barnes, A. J., Constantinou, N. C., Gibson, A. H., Kiss, A. E., Chapman, C., Reilly, J., Bhagtani, D., & Yang, L. (2024). regional-mom6: A Python package for automatic generation of regional configurations for the Modular Ocean Model 6. *Journal of Open Source Software*, 9(100), 6857. <https://doi.org/10.21105/joss.06857>
- Constantinou, N. C., Hogg, A. McC., Gibson, A., Steketee, A., Beucher, R., Proft, M., Heerdegen, A., Neme, J., Holmes, R. M., Morrison, A., Fierro-Arcos, D., Kiss, A. E., Oliveira, M., Yung, C., Doddridge, E., Huneke, W., Bhagtani, D., Ong, E. Q. Y., Munroe, J., ... Narayanan, A. (2026). *COSIMA cookbook: A cookbook of recipes for analysing ocean and sea ice model output*. <https://doi.org/10.5281/zenodo.14353852>
- Dask Development Team. (2016). *Dask: Library for dynamic task scheduling*. <http://dask.pydata.org>
- Delandmeter, P., & Seville, E. van. (2019). The parcels v2.0 lagrangian framework: New field interpolation schemes. *Geoscientific Model Development*, 12(8), 3571–3584. <https://doi.org/10.5194/gmd-12-3571-2019>
- Elson, P., Andrade, E. S. de, Lucas, G., May, R., Hattersley, R., Campbell, E., Comer, R., Dawson, A., Little, B., Raynaud, S., scm72, Snow, A. D., Igolston, Blay, B., Killick, P., lbdreyer, Peglar, P., Wilson, N., Andrew, ... Kirkham, D. (2024). *SciTools/cartopy: v0.24.1* (Version v0.24.1). Zenodo. <https://doi.org/10.5281/zenodo.13905945>
- Hoyer, S., & Hamman, J. J. (2017). xarray: N-D labeled arrays and datasets in Python. *Journal of Open Research Software*, 5(1), 10. <https://doi.org/10.5334/jors.148>
- Intake developers. (2026). *Intake: A lightweight package for finding, investigating, loading and disseminating data*. <https://intake.readthedocs.io/en/reader/>
- Kiss, A. E., Woodward, M. J., Hollings, T., Mu, M., Fiedler, R., Steven, A. D. L., Lenton, A., Matear, R., Chamberlain, M. A., Oke, P. R., Alves, O., Griffies, S. M., Hill, C., Kovach, A., Marsland, S. J., Fiedler, R., Waters, J., Yang, C.-H., Zhou, X., ... Hogg, A. McC. (2020). ACCESS-OM2 v1.0: A global ocean-sea ice model at three resolutions. *Geoscientific Model Development*, 13(2), 401–442. <https://doi.org/10.5194/gmd-13-401-2020>
- Kluyver, T., Ragan-Kelley, B., Pérez, F., Granger, B., Bussonnier, M., Frederic, J., Kelley, K., Hamrick, J., Grout, J., Corlay, S., Ivanov, P., Avila, D., Abdalla, S., & Willing, C. (2016). Jupyter notebooks—a publishing format for reproducible computational workflows. In F. Loizides & B. Schmidt (Eds.), *Positioning and power in academic publishing: Players, agents and agendas* (pp. 87–90). IOS Press. <https://doi.org/10.3233/978-1-61499-649-1-87>
- Rocklin, M. (2015). Dask: Parallel computation with blocked algorithms and task scheduling. In K. Huff & J. Bergstra (Eds.), *Proceedings of the 14th python in science conference* (pp. 130–136). <https://doi.org/10.25080/Majora-7b98e3ed-013>
- Zhuang, J., Dussin, R., Bourgault, P., Huard, D., Banihirwe, A., Raynaud, S., Malevich, B., Schupfner, M., Gauthier, C., Filipe, Levang, S., Almansi, M., Jüling, A., RichardScottOZ, RondeauG, Rasp, S., Smith, T. J., Meyer, A. G., Mares, B., ... Li, X. (2025). *pangeo-data/xESMF: v0.9.2*. <https://doi.org/10.5281/zenodo.4294774>